

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Shaik Ashraf/192324088 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____



Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

Question 1

Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Answer

A retail business can collect valuable information from billing counters, websites, mobile applications, loyalty programmes, customer support and social channels. The main problem is not the lack of data; it is that the information is separated across systems and therefore cannot be used as one consistent customer view. A unified architecture should bring these streams together, resolve customer identities, process events quickly and expose trusted information to analytics and machine-learning applications.

Proposed architecture

The recommended design follows a data-lake and streaming approach. Transactional and behavioural events are first collected through a common ingestion layer. The raw data is retained in scalable object storage, while identity-resolution services connect records belonging to the same customer.

Architecture components

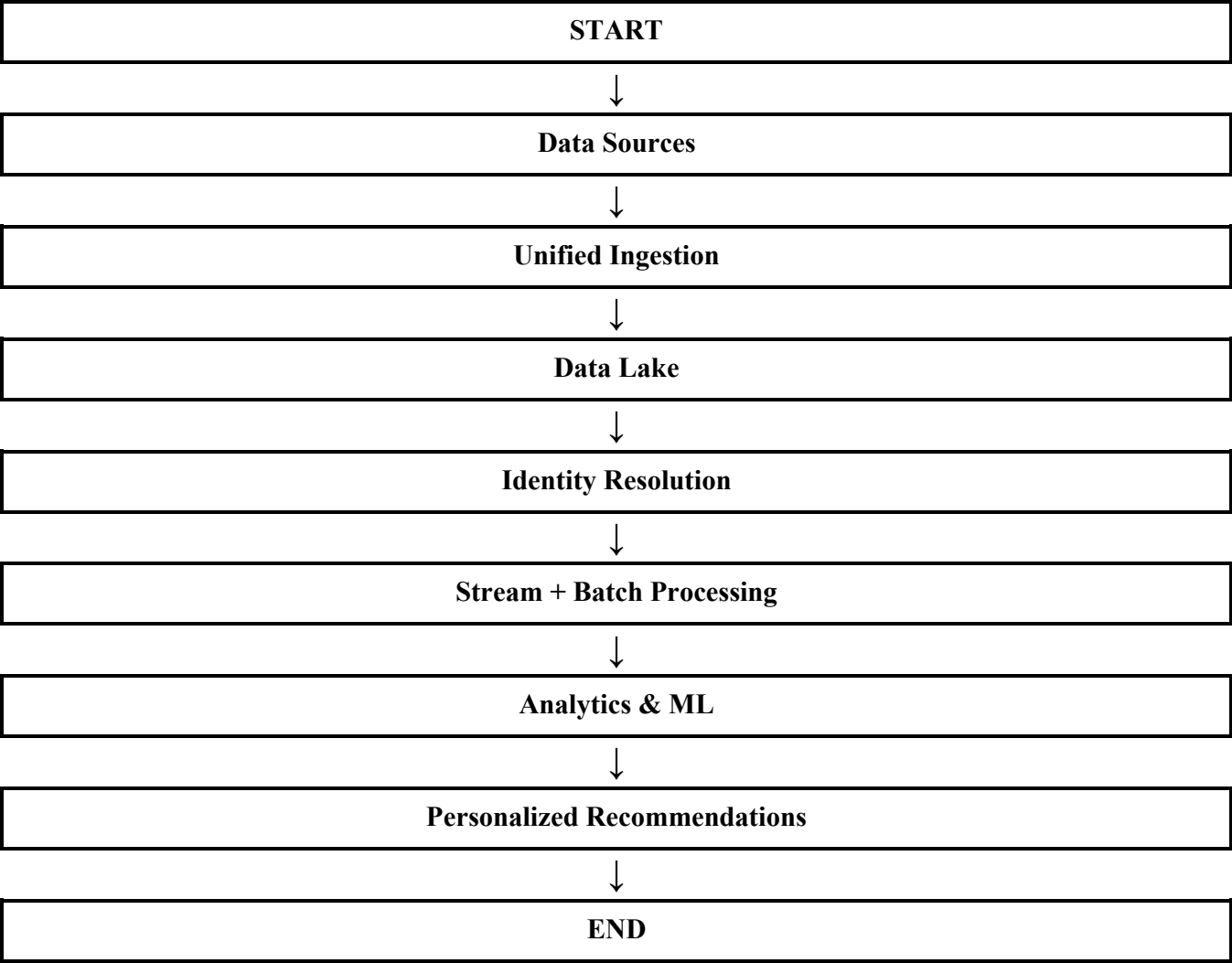
- Data ingestion: Apache Kafka or AWS Kinesis can capture point-of-sale transactions, clickstream events, application activity and CRM updates.
- Storage: Amazon S3, Azure Data Lake or HDFS can retain structured, semi-structured and unstructured data at scale.
- Identity resolution: Master Data Management and entity-matching techniques can create a consistent customer profile.
- Processing: Apache Spark and Spark Streaming or Apache Flink can support both historical and near-real-time processing.

Business outcome

Table 1 — Unified Big Data Architecture for Retail

Layer	Technology / Approach	Purpose
Ingestion	Apache Kafka / AWS Kinesis, Flume	Capture clickstream, POS transactions, app events and CRM feeds in real time
Storage (Data Lake)	HDFS / Amazon S3 / Azure Data Lake	Store raw structured, semi-structured and unstructured data cheaply at scale
Identity Resolution	Master Data Management (MDM), entity matching	Merge duplicate customer IDs across channels into one profile
Processing	Apache Spark (batch) + Spark Streaming / Flink (real-time)	Clean, transform and enrich data for both historical and live use cases
Serving / Warehouse	Snowflake / Redshift / BigQuery	Curated, query-ready data for BI and reporting
Analytics & ML	Spark MLlib, TensorFlow	Build recommendation and churn-prediction models
Visualization	Power BI / Tableau	Present unified customer insights to business teams

Question 1 — Flowchart



Question 2

Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Answer

Scientific applications such as genome analysis, climate modelling and particle-physics experiments generate datasets that can reach terabytes or exabytes. Processing all of this information on a single machine creates limits in memory, CPU capacity, storage throughput and fault recovery. Distributed computing addresses these limits by dividing the workload among several machines and executing independent tasks simultaneously.

Centralized versus distributed processing

A centralized system depends mainly on vertical scaling: the existing machine must be made more powerful. A distributed model uses horizontal scaling, where additional nodes are added as the workload increases. Hadoop MapReduce and Apache Spark are common examples of this approach.

Comparison

- Scalability: centralized systems are constrained by the capacity of one machine, whereas clusters can grow by adding nodes.
- Processing speed: independent data partitions can be processed simultaneously, reducing total execution time.
- Fault tolerance: distributed frameworks can replicate data and restart failed tasks instead of losing the complete job.
- Cost efficiency: clusters can use commodity or cloud resources rather than relying only on highly specialized hardware.

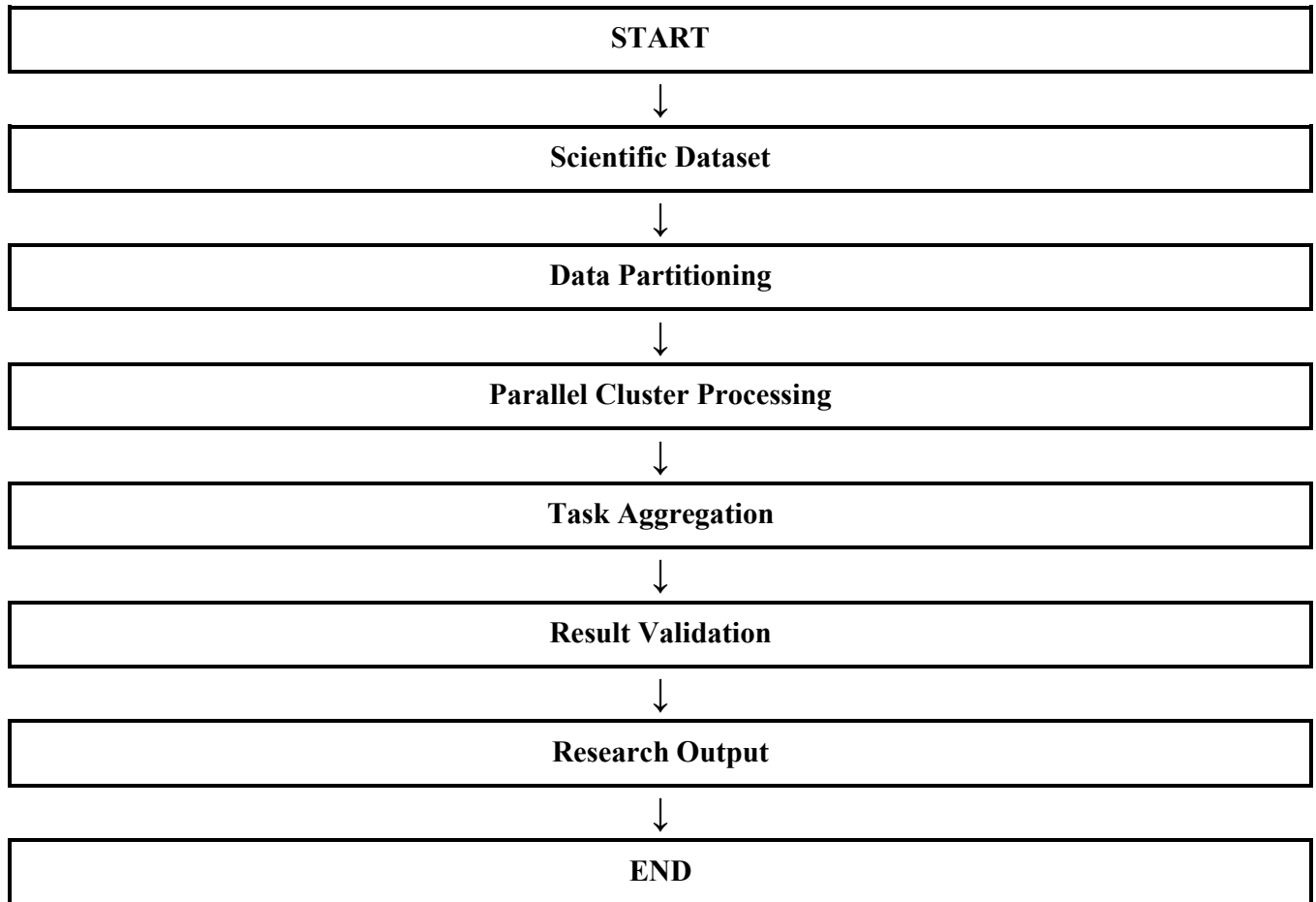
Example

Table 2 — Centralized and Distributed Processing Comparison

Parameter	Centralized System	Distributed / Parallel Model
Scalability	Vertical only (add CPU/RAM to one machine)	Horizontal — add more nodes as data grows
Fault tolerance	Single point of failure	Data replication and task re-execution on node failure
Throughput	Limited by one machine's I/O	Aggregate I/O across all nodes in the cluster
Cost	Expensive specialized hardware	Commodity hardware, cost-effective at scale
Processing time	Grows linearly with data size	Reduced via data-parallel execution across nodes
Typical use	Small/medium, latency-sensitive jobs	Petabyte-scale batch and iterative scientific workloads

In genome sequencing, millions of reads can be divided into groups and processed by different nodes at the same time. Instead of waiting for one machine to complete the entire workload, the cluster performs many alignments concurrently. This makes distributed processing particularly effective when both dataset size and computational complexity are high.

Question 2 — Flowchart



Question 3

Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Answer

A digital payment platform cannot depend only on end-of-day reports because fraudulent transactions may cause damage within seconds. A streaming-first architecture is therefore appropriate: each transaction is evaluated as it arrives, behavioural features are calculated immediately, and a machine-learning model produces a risk score before the payment is finalized.

Proposed workflow

- Event capture: Kafka or Kinesis receives transaction details such as amount, merchant, location, device information and timestamp.
- Feature generation: Spark Streaming or Flink calculates short-term behavioural indicators such as transaction frequency, spending deviation and geographic distance.
- Risk scoring: Random Forest, XGBoost or a neural network estimates the probability that a transaction is fraudulent.
- Decision layer: a rules engine combines the model score with business rules and chooses approve, review or block.
- Feedback and retraining: confirmed fraud outcomes are stored and used to refresh the model regularly.

Why streaming is effective

Fraud often appears as a sequence rather than a single suspicious transaction. For example, a sudden series of transactions from unusual locations may be harmless individually but highly suspicious when considered together. Streaming analytics makes these relationships visible immediately. Predictive models further improve detection by learning combinations of signals that are difficult to capture through fixed rules.

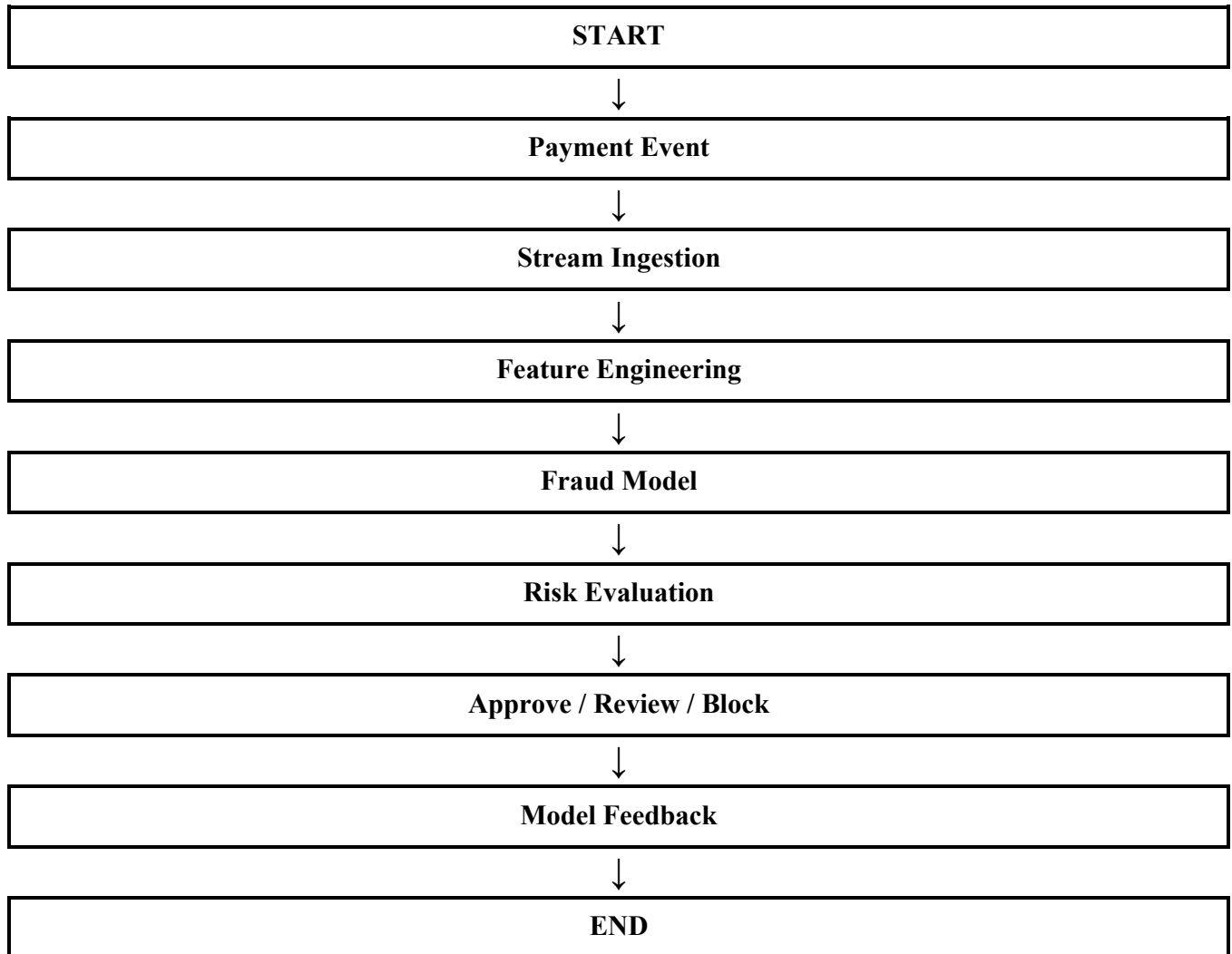
Expected outcome

Table 3 — Real-Time Fraud Detection Framework

Stage	Component	Function
1. Ingestion	Kafka / Kinesis stream	Capture every transaction event in real time
2. Feature engineering	Spark Streaming / Flink	Compute rolling features: transaction velocity, average spend, geo-distance from last transaction
3. Scoring	ML model (Random Forest / XGBoost / neural net)	Assign a real-time fraud-probability score
4. Decision	Rules engine + threshold	Approve, flag for review, or block instantly
5. Feedback loop	Batch retraining (Spark MLlib)	Retrain the model nightly using confirmed fraud/non-fraud outcomes

The combination of low-latency processing, behavioural features and model-based scoring can stop suspicious payments earlier while reducing unnecessary blocking of genuine customers. A tiered decision process also helps balance security with customer experience.

Question 3 — Flowchart



Question 4

Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Answer

Multimedia data is fundamentally different from conventional tables because the files are large, heterogeneous and difficult to search directly. A single video may contain valuable information that is not visible from its filename or basic metadata. A suitable architecture should therefore separate large raw files from the structured information extracted from them.

Major challenges

- Storage growth: high-resolution video and image collections can consume very large amounts of storage.
- Format diversity: systems must handle formats such as MP4, JPEG, WAV and RAW without assuming a single schema.
- Processing demand: transcoding, speech recognition and computer-vision inference require substantial CPU or GPU resources.
- Searchability: raw binary files cannot be queried as easily as structured records.
- Network cost: moving large media files repeatedly increases bandwidth usage and latency.

Recommended technology stack

- Use Amazon S3, Azure Blob Storage or HDFS as scalable bulk storage for raw media.
- Adopt a data-lake model with schema-on-read so different media formats can coexist.
- Use GPU-enabled processing pipelines for transcoding, computer vision and speech-to-text workloads.

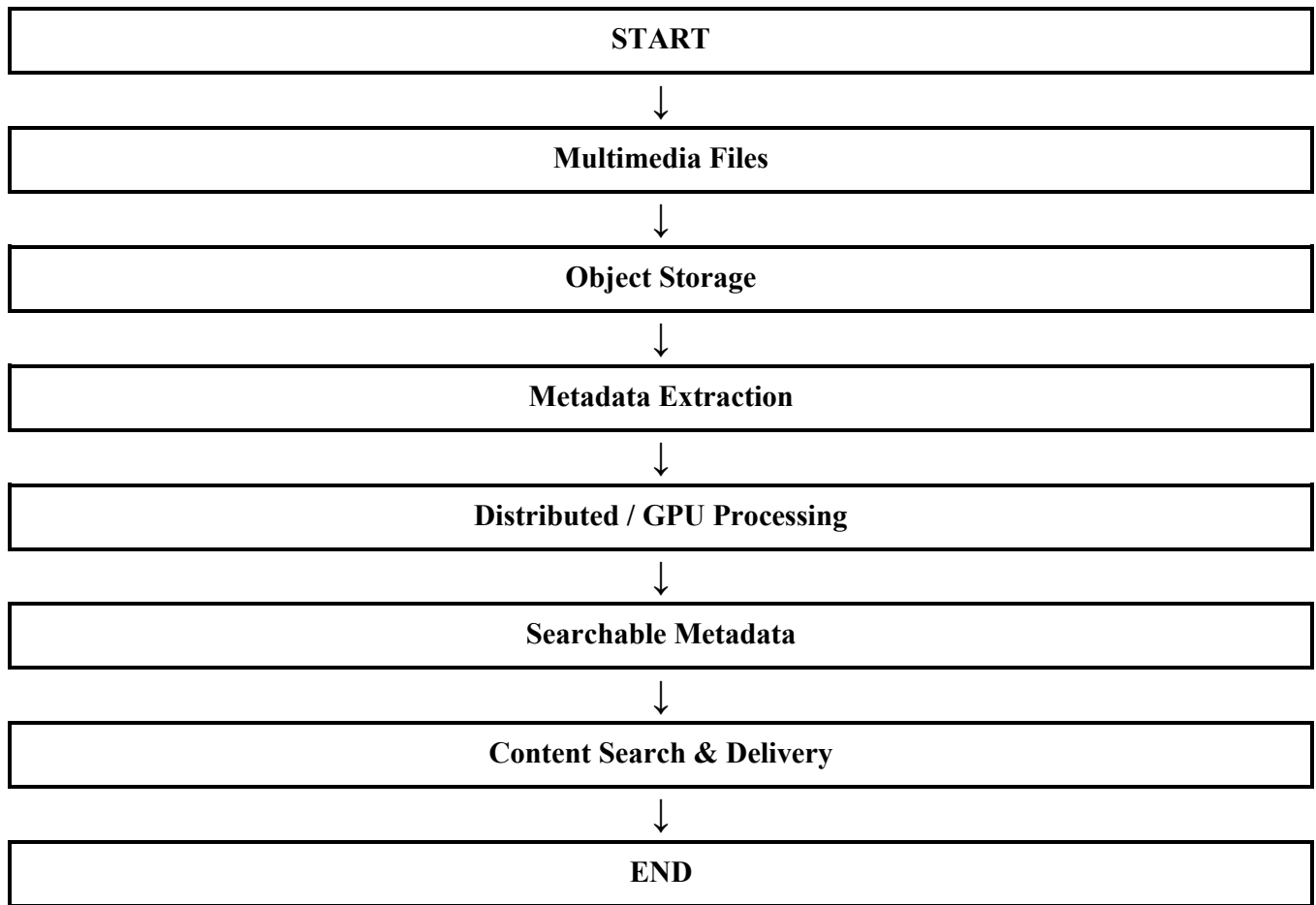
Design principle

Table 4 — Multimedia Data Challenges and Technology Choices

Challenge	Why It Matters	Recommended Technology
Storage volume	A single hour of 4K video can exceed 40 GB	Object storage — Amazon S3, Azure Blob, HDFS with erasure coding
Format heterogeneity	No fixed schema across audio/video/image types	Data lake architecture (schema-on-read)
Processing cost	Transcoding and inference are compute-heavy	GPU-accelerated clusters, Spark with distributed video-processing libraries
Content search	Cannot query raw bytes directly	Metadata tagging + deep-learning feature extraction (CNNs) for indexing
Bandwidth for transfer	Large files slow ingestion and delivery	CDN caching and edge storage close to users
Retention & cost control	Not all content is accessed equally often	Tiered/cold storage (e.g., S3 Glacier) for infrequently accessed media

The most practical approach is to keep raw multimedia in economical object storage and maintain a separate, lightweight metadata layer for search and analytics. This reduces query cost while preserving the original content for future processing.

Question 4 — Flowchart



Question 5

Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

Answer

A cloud analytics platform should be designed on the assumption that failures will occur. The recovery plan must therefore address local hardware failures, availability-zone or regional outages, accidental deletion and application errors. The objective is not only to protect data but also to restore critical services within an agreed recovery time.

Recovery architecture

- Redundancy: maintain multiple copies of important data across nodes or availability zones.
- Geo-replication: replicate critical datasets to another region so a regional outage does not destroy the recovery option.
- Automated failover: use health checks and load balancing to redirect applications to a standby environment.
- Backups: combine scheduled snapshots with incremental backups to support point-in-time recovery.
- Monitoring: use centralized monitoring and alerts to detect failures early.
- Testing: conduct periodic recovery drills so that documented procedures are verified in practice.

Resilience measures

- Redundancy reduces the effect of individual disk or node failures.
- Cross-region replication provides protection against large-scale infrastructure failures.
- Automated failover reduces service interruption and supports a lower Recovery Time Objective (RTO).
- Backups help achieve a lower Recovery Point Objective (RPO) when data is corrupted or deleted.
- Regular disaster-recovery tests expose configuration or dependency problems.

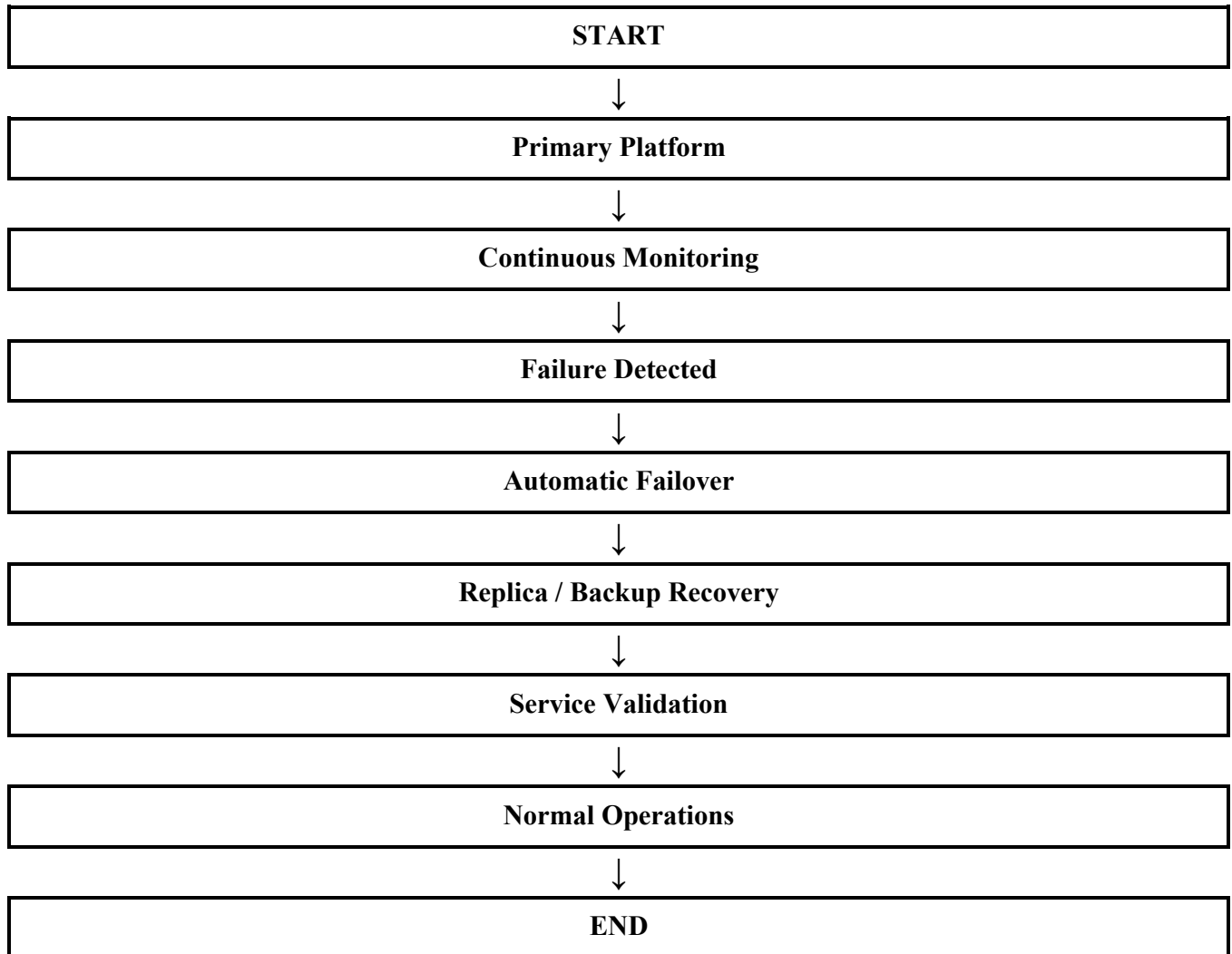
Conclusion

Table 5 — Disaster Recovery and Business Continuity Strategy

Component	Mechanism	Resilience Benefit
Data redundancy	3x replication in HDFS / multi-AZ storage	Survives individual disk or node failure with no data loss
Geo-replication	Cross-region replication (e.g., S3 Cross-Region Replication)	Protects against full regional outage or natural disaster
Automated failover	Load balancer + health checks routing to standby cluster	Minimizes downtime (low RTO) without manual intervention
Backup strategy	Scheduled snapshots + incremental backups	Enables point-in-time recovery (low RPO) after corruption or error
DR testing	Periodic failover drills / chaos engineering	Validates that recovery procedures actually work under load
Monitoring & alerting	Centralized observability (Prometheus/CloudWatch)	Detects failures early, triggering failover before user impact

A strong DR plan combines prevention, detection and recovery. Replication protects information, automated failover protects availability, and tested backups protect against logical errors. Together, these controls provide much stronger resilience than relying on a single backup mechanism.

Question 5 — Flowchart



Question 6

Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Answer

At large scale, small data-quality problems can become major analytical errors because the same records are reused across dashboards, models and business processes. Inconsistent formats can produce incorrect joins, duplicate records can inflate key performance indicators, and missing values can distort statistical analysis. Data governance should therefore be treated as an ongoing operational process rather than a one-time cleaning exercise.

Quality problems and their effects

- Inconsistency: different date, currency or measurement formats can produce incorrect comparisons and aggregations.
- Duplication: repeated customer or transaction records can exaggerate counts, revenue and other KPIs.
- Missing values: incomplete records can reduce sample size or introduce bias into machine-learning models.
- Stale data: delayed refreshes can lead decision-makers to act on information that no longer represents current conditions.
- Poor traceability: without lineage, analysts may struggle to identify the source of an incorrect value.

Governance controls

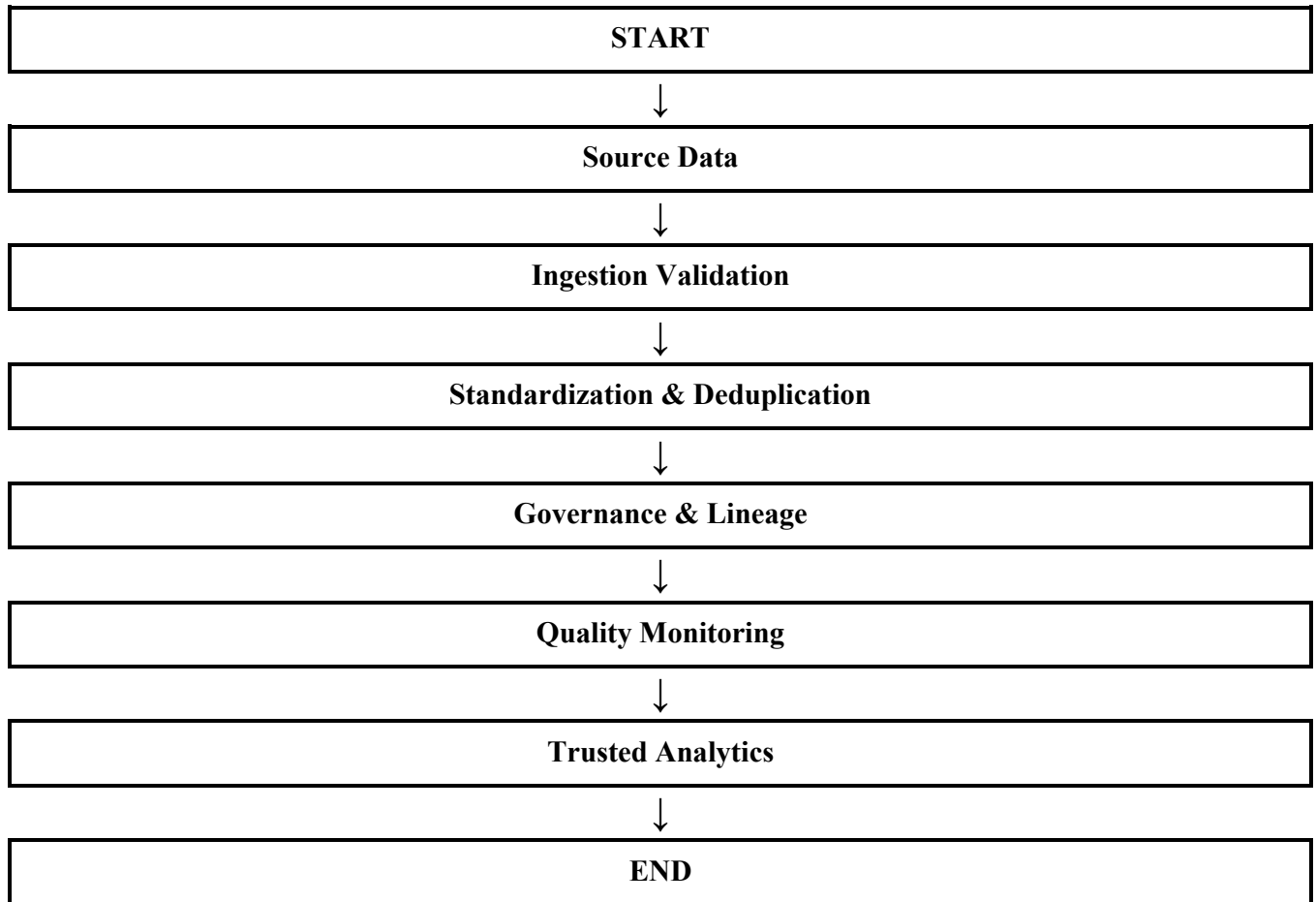
- Apply validation rules at the point of ingestion instead of waiting until reports are generated.
- Maintain common schemas, business definitions and data dictionaries across departments.
- Use entity resolution and deduplication rules for important master data.
- Document approved methods for handling missing values and exceptions.
- Assign data stewards to major domains such as sales, finance and customer data.
- Maintain metadata and lineage so every important metric can be traced back to its source.
- Monitor completeness, consistency and duplication rates through automated quality dashboards.

Result

Table 6 — Data Quality Issues and Governance Controls

Issue	Typical Impact	Governance Strategy
Inconsistency	Incorrect joins and aggregations across sources	Enforce standardized schemas and data dictionaries at ingestion
Duplication	Inflated counts, skewed KPIs	Deduplication rules and entity-resolution / master data management
Missing values	Biased models, reduced sample size	Validation rules at capture point + documented imputation policy
Stale data	Decisions based on outdated snapshots	Defined data freshness SLAs and automated refresh pipelines
No ownership	No accountability when errors are found	Assign data stewards per domain (sales, customer, finance)
Poor traceability	Cannot audit where a number originated	Data lineage tracking and metadata catalogs

Question 6 — Flowchart



Question 7

Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Answer

A streaming platform receives millions of behavioural events such as plays, pauses, skips, search queries, ratings and viewing duration. These events provide a richer picture of user preference than occasional ratings alone. The recommendation system should combine real-time features with large-scale historical learning so that recommendations remain relevant as user interests change.

System design

- Event collection: Kafka or Kinesis captures viewing and interaction events.
- Feature management: Spark Streaming and a fast feature store such as Redis maintain current user and content features.
- Model training: Spark MLlib with algorithms such as ALS can learn collaborative preferences from historical interactions.
- Content intelligence: metadata embeddings can represent genre, cast, language and other title attributes.
- Real-time serving: a low-latency model-serving layer ranks candidate titles for each request.
- Experimentation: A/B testing measures click-through, watch time and retention to guide improvements.

Why this approach works at scale

Collaborative filtering involves a large user-item matrix. Distributed matrix factorization allows the training workload to be divided across many machines. At the same time, implicit signals such as watch duration and skip behaviour provide continuous evidence of preference, allowing the model to adapt without waiting for users to submit explicit ratings.

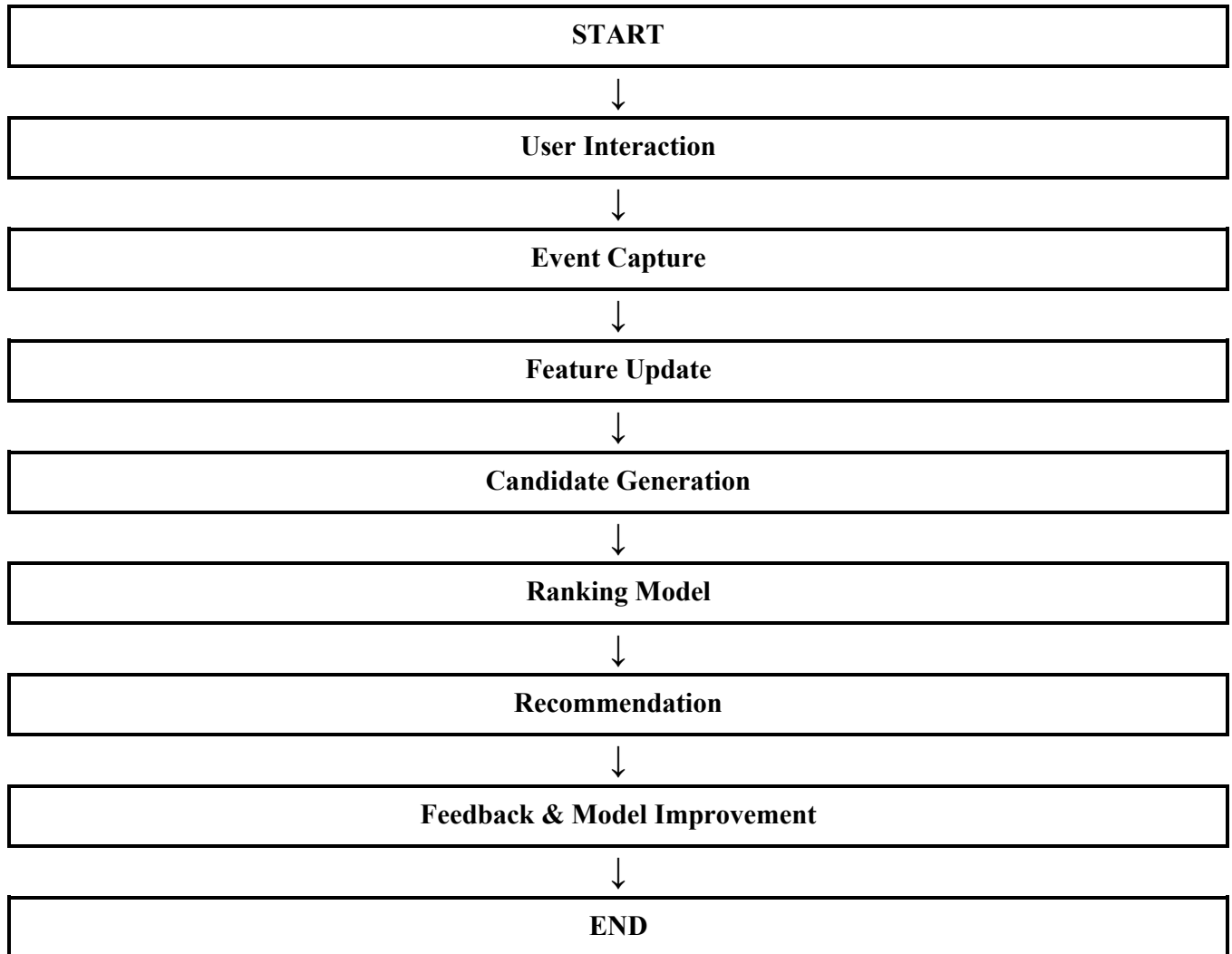
Business impact

Table 7 — Scalable Recommendation System Components

Component	Technology	Role
Event capture	Kafka / Kinesis	Stream every play, skip, pause and rating event
Feature store	Spark Streaming + Redis	Maintain up-to-date user and content feature vectors
Batch model training	Spark MLlib (ALS collaborative filtering)	Learn latent user-item preference patterns from historical data
Real-time model serving	TensorFlow Serving / low-latency API	Return ranked recommendations within milliseconds of a request
Content-based layer	NLP / embeddings on metadata (genre, cast, tags)	Handle cold-start for new titles with little interaction history
A/B testing	Experimentation platform	Continuously measure and improve recommendation click-through

Better recommendations can increase relevant viewing, reduce search effort and improve retention. Content-based features also reduce the cold-start problem for newly added titles that have little historical interaction data.

Question 7 — Flowchart



Question 8

Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Answer

Cloud computing has become a practical foundation for big data because organizations can provision storage and compute resources when they are needed instead of maintaining a fixed infrastructure. Managed services such as Amazon EMR, Azure HDInsight and Google Dataproc can reduce the operational effort required to build large analytics clusters. However, cloud adoption still requires disciplined cost, security and architecture management.

Advantages

- Elastic scalability allows resources to increase or decrease with workload demand.
- Pay-as-you-go pricing can reduce the need for large upfront infrastructure investment.
- Managed services shorten deployment time and reduce routine administration.
- Cloud platforms provide broad choices for storage, streaming, databases and machine learning.
- Built-in security services, encryption options and compliance capabilities can strengthen the overall security posture.

Limitations and controls

- Uncontrolled clusters, storage and data-transfer charges can create cost sprawl; budgets, tagging and automated shutdown policies help control this.
- Provider-specific services may create vendor lock-in; open formats such as Parquet and portable architecture reduce the risk.

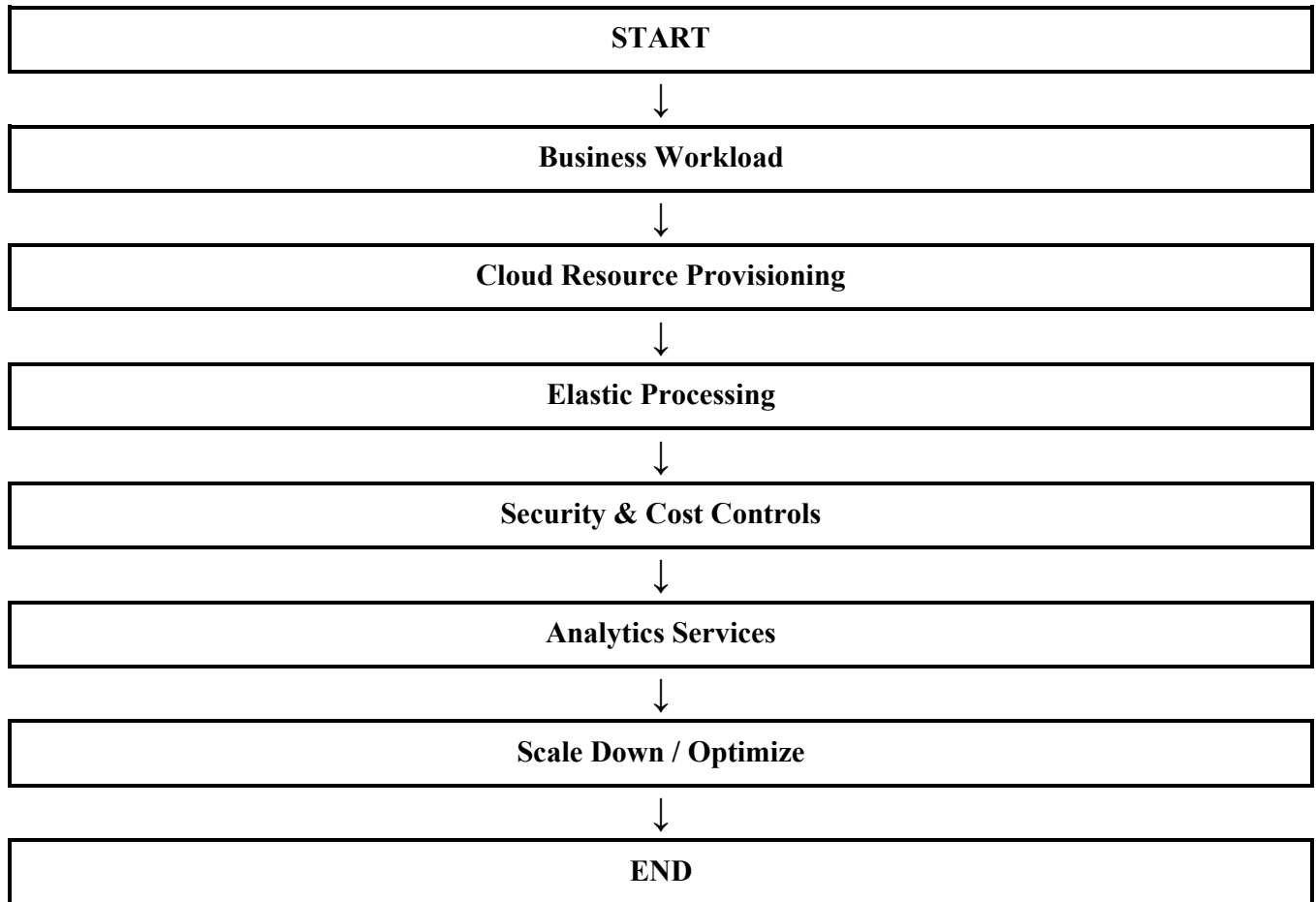
Conclusion

Table 8 — Cloud Computing: Advantages and Limitations

Dimension	Advantage	Limitation
Scalability	Elastic scaling up/down within minutes to match workload	Sudden scale-up can hit provider quota or regional capacity limits
Cost management	Pay-as-you-go avoids large upfront capital expenditure	Costs can spiral without governance (idle clusters, data egress fees)
Flexibility	Wide choice of managed services (storage, ML, streaming)	Risk of vendor lock-in when using provider-specific APIs
Security	Provider-grade physical security, encryption, compliance certifications	Shared responsibility model — misconfiguration by the customer remains a major breach cause
Speed of deployment	Clusters provisioned in minutes vs. weeks for on-prem hardware	Data transfer to/from cloud can be slow and costly for very large datasets

Cloud computing is particularly valuable when workloads are variable or rapidly growing. Its benefits are strongest when organizations combine elastic infrastructure with financial governance, security controls and portable data design.

Question 8 — Flowchart



Question 9

Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Answer

Modern logistics systems generate continuous information from GPS devices, vehicle sensors, traffic services, weather feeds and order systems. Instead of analysing these sources only after a trip, a real-time platform can turn the incoming data into operational decisions while vehicles are still on the road.

Framework

- Vehicle telemetry and IoT sensors provide location, speed, fuel and engine information.
- Kafka or another streaming platform transports events to the analytics pipeline.
- Spark Streaming and graph-based algorithms evaluate road conditions and calculate alternative routes.
- Historical telematics data supports fuel-efficiency analysis and driver-behaviour studies.
- Machine-learning models predict maintenance requirements from engine and vibration signals.
- Time-series forecasting helps estimate future demand and allocate vehicles and warehouse resources.
- Power BI or Tableau dashboards give dispatch teams a live operational view.

Operational benefits

Table 9 — Real-Time Transportation and Logistics Analytics

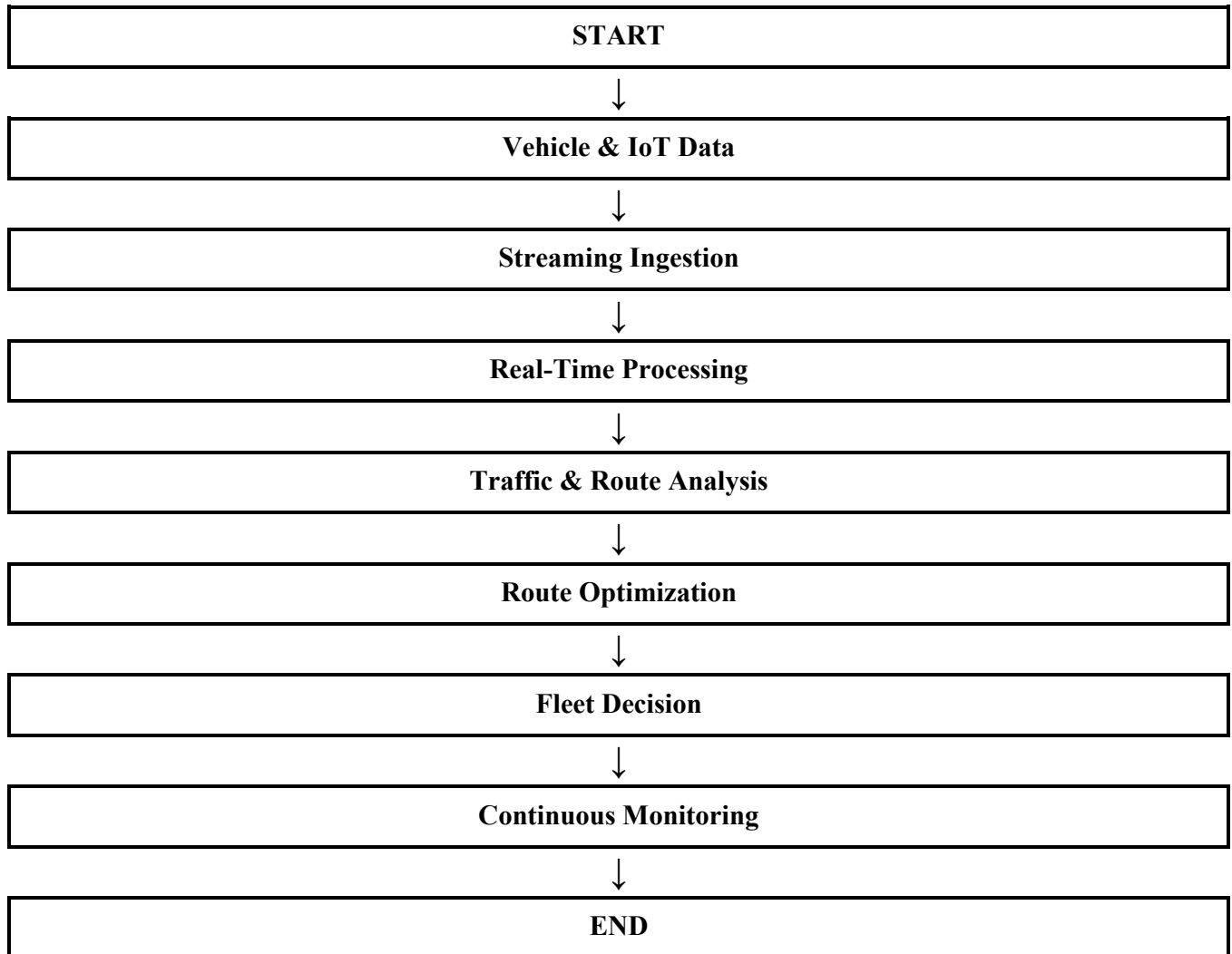
Use Case	Big Data Technology	Operational Benefit
Live vehicle tracking	IoT sensors + Kafka streaming ingestion	Continuous visibility of fleet location and status
Dynamic route optimization	Spark Streaming + graph algorithms	Re-routes around traffic/accidents in real time
Fuel efficiency analysis	Batch analytics on historical telematics (Spark/Hadoop)	Identifies inefficient driving patterns and idle time
Predictive maintenance	ML models on sensor vibration/engine data	Flags vehicles needing service before breakdown
Demand forecasting	Time-series models on order history	Improves fleet allocation and warehouse staffing
Dashboarding	Power BI / Tableau real-time dashboards	Gives dispatchers a live operational overview

- Dynamic route selection can avoid congestion, accidents and unnecessary idling.
- Fuel analysis identifies inefficient driving patterns and high-consumption routes.
- Predictive maintenance can identify vehicles that require attention before a breakdown occurs.
- Demand forecasting improves fleet allocation and warehouse staffing.
- Live dashboards allow dispatchers to respond to changes without waiting for a nightly report.

Conclusion

The main advantage of big data in logistics is the ability to connect large volumes of sensor and operational information with decisions made in seconds. This improves visibility and can reduce avoidable fuel consumption, downtime and delivery delays.

Question 9 — Flowchart



Question 10

Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Answer

Big data can create significant value, but collecting large amounts of personal information also increases the consequences of misuse. Organizations must consider consent, fairness, privacy, security and regulatory obligations throughout the data lifecycle. Responsible analytics therefore requires a combination of technical safeguards, governance processes and transparent communication.

Key ethical and legal concerns

- Informed consent: information collected for one purpose may later be reused in ways users did not expect.
- Algorithmic bias: models trained on unbalanced data can produce systematically unfair outcomes.
- Re-identification: supposedly anonymous datasets may still reveal individuals when combined with other information.
- Cross-border compliance: organizations operating in multiple regions may face different privacy and data-residency requirements.
- Data breaches: large centralized datasets can become high-value targets for attackers.
- Transparency: users may not understand what is collected, how it is analysed or how long it is retained.

Recommended organizational policies

- Collect only the information required for a defined purpose and obtain clear, granular consent where required.
- Perform periodic fairness and bias assessments for models that affect people.
- Apply strong access control, encryption, pseudonymization and appropriate anonymization techniques.
- Maintain a data-residency and regulatory mapping process for each operating region.
- Prepare an incident-response and breach-notification procedure and test it regularly.
- Publish a clear privacy policy explaining data collection, usage and retention.
- Maintain accountable privacy and data-governance roles, including a Data Protection Officer where applicable.

Final assessment

Table 10 — Ethical and Legal Challenges with Policy Responses

Challenge	Concern	Recommended Policy
Informed consent	Data reused beyond its original purpose	Purpose limitation policy with explicit, granular consent
Algorithmic bias	Models discriminate against protected groups	Mandatory bias audits and diverse training data reviews
Re-identification risk	"Anonymized" data can be de-anonymized	Strong anonymization/pseudonymization standards and access controls
Cross-border compliance	Conflicting requirements (GDPR, CCPA, DPDP)	Data residency mapping and a dedicated compliance/legal review process
Data breach exposure	Loss of sensitive personal data	Encryption at rest/in transit, incident-response plan, breach-notification protocol
Lack of transparency	Users don't know how their data is used	Public privacy policy, data-usage dashboards, and a Data Protection Officer role

Overall Key Takeaways

- Big data architectures should combine scalable storage, distributed processing and appropriate real-time components according to the business requirement.
- Data quality and governance are as important as computing technology because unreliable data leads directly to unreliable analytics.
- Streaming analytics is essential when decisions must be made within seconds, such as fraud detection, fleet management and personalization.
- Cloud and distributed platforms provide scale and flexibility, but they require cost controls, security practices and careful architectural design.
- Responsible use of big data requires privacy, fairness, transparency, accountability and compliance to be built into the data lifecycle.

Question 10 — Flowchart

